ANH TONG

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SUMMARY

Data science graduate who builds tested data pipelines and machine learning systems in Python and SQL. Projects include a layered SQL pipeline with automated data quality checks feeding a leakage-safe prediction model, a real-time streaming fraud detection platform, and a team-built RAG chatbot. Looking to bring reliable data engineering and applied AI to Deloitte's Government & Public Services clients.

EDUCATION

University of Utah — Salt Lake City, UT **Aug 2021 – Dec 2025**
B.S. in Data Science | GPA: 3.556 / 4.0 | Dean's List: Fall 2021, Spring 2024, Spring 2025

SKILLS

Data Engineering: SQL, Python (pandas, NumPy), DuckDB, PostgreSQL, ETL pipelines, data quality testing, Spark Structured Streaming, Kafka/Redpanda, Feast
Machine Learning: scikit-learn, XGBoost, TensorFlow, MLflow, cross-validation, logistic regression
AI / LLM: LangChain, OpenAI API, Pinecone, RAG, vector search
Deployment & Tools: FastAPI, Docker, Git, CI/CD

PROJECTS

Appointment Attendance Pipeline & Model | SQL Data Pipeline & Risk Analysis **Oct 2026**

- Built a three-layer pipeline (raw, staging, features) ~110K medical appointments that rebuilds from source with one command, with zero row loss through staging.
- Wrote 16 data quality checks that run after each layer and halt the build on failure. A test suite corrupts the database 10 ways to confirm each check fires.
- Found and fixed silent data bugs: a timezone cast that shifted every date back one day, and 15-digit patient IDs losing precision as floats.
- Built leakage-safe patient history features with a SQL self-join, and split train/test by patient since 66% of rows were repeat visits.
- Trained a logistic regression (71 features cut to 17, AUC 0.724). The top 5% of predicted risk no-showed at 2x the base rate, holding on both unseen patients and a later time period.

ShopSentry | Real-Time Anomaly Detection Platform **May 2026**

- Built a streaming pipeline (Redpanda, Spark Structured Streaming, Feast) from simulated transaction intake to model scoring.
- Compared 5 detection approaches with nested cross-validation, then selected a TensorFlow autoencoder and XGBoost setup (F1 = 0.84).
- Served models through FastAPI with MLflow tracking. Load-tested at 32ms median latency with 20 concurrent users, over 30,000+ requests with zero failures.
- Audited the system and documented its limits (synthetic data, a leaky labeler, train/serve preprocessing skew) along with the fixes each would need.

Cooked | AI-Powered Recipe Discovery App | Capstone **Jan – Dec 2025**

- Led AI and data engineering on a 4-person team building a recipe app, owning search and the chatbot over 3K+ recipes.
- Built a RAG chatbot (LangChain, OpenAI, Pinecone) that embeds user requests, retrieves the closest real recipes, and returns ranked matches in a conversational reply.
- Built recipe customization for dietary needs (gluten free, dairy free, vegetarian) that edits a retrieved recipe instead of generating a new one, returning ingredient swaps, updated steps, and allergen warnings.
- Built the FastAPI backend and React Native front end connecting user queries to results.

WORK EXPERIENCE

Salt Lake County Youth Services — Salt Lake City, UT **Dec 2024 – May 2026**
Youth Leader, After School Program

- Designed and led a weekly data science curriculum for cohorts of 5 to 20 high school students, turning probability and programming into hands-on simulations for non-technical learners.
- Recruited students by tabling during lunch and got school administration to share program posts, and shifted content to video, growing Instagram views ~15x (100 to 1,500) in 30 days.
- Built trust with students from varied backgrounds, de-escalating conflicts and guiding them through personal challenges and post-graduation planning.